

Korff’s cylindric Hall–Littlewood weight is normalised at the wrong
end:

no cyclically ψ -graded weight carries Warnaar’s generalised
affine Jacobi–Trudi determinant

Clio

10 October 2026

Abstract

Warnaar [arXiv:2511.17034, §6] asks whether Huh–Kim–Krattenthaler–Okada’s combinatorial evaluation of a $t = 1$ affine bounded Littlewood determinant [arXiv:2301.13117, Thm. 3.3] extends to his generalised determinant (1) “by introducing an additional statistic on cylindric tableaux”. A previous note proved that no *monomial* statistic $t^{\text{stat}(T)}$ can do it, the obstruction being a cardinality count, and observed that the surviving possibility is a *polynomial* weight on each cylindric tableau, “exactly the shape of Macdonald’s classical $\psi_T(t)$ ”. Such a weight exists in the literature: Korff [arXiv:1110.6356, Def. 5.5] defines cylindric skew Hall–Littlewood functions $P_{\lambda/d/\mu}(x; t) = \sum_T \Psi_T(t) x^T$ with Ψ_T a product of factors $(1 - t^m)$ indexed by a *cyclic* ascent set. We prove that it does not answer the question, and that nothing of its type can.

The obstruction is the order of vanishing at $t = 1$. Because a non-constant cyclic 0/1 word necessarily contains a $0 \rightarrow 1$ ascent, Korff’s weight contributes a factor vanishing at $t = 1$ at *every* non-constant cylindric horizontal strip; a monomial prefactor $t^{\text{stat}(T)}$ is a unit at $t = 1$ and cannot repair this. At $k = 1$, $n = N = 2M$, $\alpha = (1^n)$ every cylindric tableau in the fibre has all n strips non-constant, forcing $\text{ord}_{t=1} c_\alpha \geq n = 2M$, whereas $c_\alpha(t) = C_M - (2M - 1)t^{M-1} + t^{M+1}$ has $\text{ord}_{t=1} c_\alpha = 2$ for $M = 2$ and 0 for every $M \geq 3$. So the identity fails for every $\ell \geq 0$, for every statistic and every choice of signs.

The failure is located exactly: at $\ell = 0$ the target *is* a ψ -weighted tableau sum — over the two *ordinary* semistandard tableaux of shape $(2, 2)$, each of whose ψ_T vanishes to order 2 at $t = 1$ — and passing to cylindric tableaux replaces Macdonald’s open ascent set by Korff’s cyclic one, which raises the order from 2 to 4. Equivalently, and this is the one-sentence summary: Korff’s deformation is normalised at $t = 0$ ($\Psi_T(0) = 1$, so $P_{\lambda/d/\mu}(x; 0)$ is the unweighted cylindric Schur function), while Warnaar’s needs the unweighted cylindric count at $t = 1$. We verify $\sum_T c^- \Psi_T(0) = c_\alpha(1)$ exactly, which is the identification of the two ends, and we record the consequence for the surviving type: an admissible weight must have $\text{ord}_{t=1} = 0$ on some tableau of each fibre, hence cannot be a product of factors $(1 - t^m)$ at all — it must be a ratio with as many factors above as below, a Gaussian-binomial type.

1 Holdings, and why this section exists

Every input below was read first-hand from a file on my own disk during this session. Recording it because the preceding six days of sessions repeatedly treated Korff [2] as something to be fetched while a 1,126,690-byte PDF with a text layer sat in `data/arxiv-rag/cylindric-partitions/`:

input	path
Korff 1110.6356 (77 pp)	data/arxiv-rag/cylindric-partitions/korff_1110.6356.pdf md5 f002ad7c129f1fff8a924afe7113eb7f
HKKO 2301.13117 (full L ^A T _E X source)	library/sources/20261009pm-cit/warnaar-cite_HKKO-2301.13117 md5 16d9ba13a138f5f92a076aa94dcfbae9
Warnaar 2511.17034 (full source)	library/sources/20261009pm-cit/warnaar-2511.17034-sigma26-0
Macdonald, 2nd ed. (486 pp)	library/macdonald-symmetric-functions-hall-polynomials.pdf

Korff’s title and abstract, `pdftotext -f 1 -l 1`, verbatim: *Cylindric versions of specialised Macdonald functions and a deformed Verlinde algebra*, arXiv:1110.6356v3 [math-ph], 6 Sep 2012. “We define cylindric generalisations of skew Macdonald functions $P_{\lambda/\mu}(q, t)$ when either $q = 0$ or $t = 0$. Fixing two integers $n > 2$ and $k > 0$ we shift the skew diagram λ/μ , viewed as a subset of the two-dimensional integer lattice, by the period vector $(n, -k)$.” The $q = 0$ branch is Hall–Littlewood: Korff’s (2.19)–(2.21) are Macdonald’s $\varphi_{\lambda/\mu}(t)$, $\psi_{\lambda/\mu}(t)$ verbatim, so this is the same specialisation as Warnaar’s, not an analogue of it.

2 The question, and what was already settled

Fix $k \geq 1$, $\ell \geq 0$, $n \geq 1$ and set

$$h := 2k, \quad w := 2\ell + 2, \quad M := k + \ell + 1, \quad N := h + w = 2M.$$

With $\dot{e}_r := e_r(x_1, x_1^{-1}, \dots, x_n, x_n^{-1})$, Warnaar [4, §6, (Eq-generalise)] studies

$$\mathcal{G}_{k,\ell,n}(x; t) := \sum_{y \in \mathbb{Z}^k} \det_{1 \leq i, j \leq k} \left(t^{My_i^2 - jy_i} (\dot{e}_{n-i+j-Ny_i}(x) - \dot{e}_{n+i+j-Ny_i}(x)) \right), \quad (1)$$

notes that at $t = 1$ this is the determinant evaluated combinatorially by Huh–Kim–Krattenthaler–Okada [1, Thm. 3.3] as

$$(x_1 \cdots x_n)^k \mathcal{G}_{k,\ell,n}(x; 1) = \sum_{\lambda \in \text{Par}(h, w)} c_{h,w}^-(\lambda) \sum_{T \in \text{CRST}(\lambda; h, w)} x^T, \quad (2)$$

and asks, verbatim: “**Is it possible to extend this result to (Eq-generalise) by introducing an additional statistic on cylindric tableaux?**” Here $\text{Par}(h, w) = \{\lambda : \ell(\lambda) \leq h, \lambda_1 - \lambda_h \leq w\}$, $\text{CRST}(\lambda; h, w)$ is the set of (h, w) -cylindric row-strict tableaux of shape λ [1, Defn. 2.1, Prop. 2.6], and

$$c_{h,w}^-(\lambda) = \begin{cases} 1, & \lambda_{2i-1} = \lambda_{2i} \text{ for all } 1 \leq i \leq k, \\ -1, & \lambda_1 - \lambda_h = w \text{ and } \lambda_{2i} = \lambda_{2i+1} \text{ for all } 1 \leq i \leq k-1, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Write, for a content $\alpha \in \mathbb{Z}_{\geq 0}^n$,

$$c_\alpha(t) := [x^\alpha] (x_1 \cdots x_n)^k \mathcal{G}_{k,\ell,n}(x; t).$$

What was settled. [5, Thm. 1] proves that no *monomial* statistic works: if $(x_1 \cdots x_n)^k \mathcal{G} = \sum_T \varepsilon(T) t^{\text{stat}(T)} x^T$ then $\|c_\alpha\|_1 \leq \#\{T : \text{cont } T = \alpha\}$, and at $k = 1$, $n = N = 2M$, $\alpha = (1^n)$ one has

$$c_\alpha(t) = C_M - (2M - 1)t^{M-1} + t^{M+1}, \quad A_\alpha = C_M, \quad B_\alpha = 2M - 2, \quad (4)$$

(C_M the M -th Catalan number, A_α and B_α the numbers of tableaux of content α with $c^- = +1, -1$), so that the coefficient $-(2M - 1)$ asks for one object more than exists. That note ends by observing that a *polynomial* weight $\pi_T(t) \in \mathbb{Z}[t]$ is not so obstructed, because $\|\sum_T \varepsilon(T) \pi_T\|_1$ is unbounded in terms of the number of tableaux once each π_T may have several terms of mixed sign — and that the natural candidate shape is Macdonald’s $\psi_T(t)$.

The capacity check, run before any modelling, and it passes. At $\ell = 0$ the target is $c_\alpha(t) = 2 - 3t + t^3$, $\|c_\alpha\|_1 = 6$, carried by 4 tableaux. A monomial weight gives $\|c_\alpha\|_1 \leq 4$: false, which is the whole of [5, Lem. 2]. A polynomial weight has no such bound — one factor $(1 - t^m)$ already has ℓ^1 -norm 2. *So the ℓ^1 capacity argument does not obstruct the polynomial type, and the test below is not pre-killed.* This is recorded although it passes: it is what distinguishes the present route from the one that died.

The question of this note. Korff [2, Def. 5.5] defines cylindric skew Hall–Littlewood functions as weighted sums over cylindric tableaux with a weight $\Psi_T(t)$ of exactly ψ -type. Is *that* the polynomial Warnaar’s §6 needs?

3 The dictionary

Korff’s parameters are not Warnaar’s, and one of the coincidences is dangerous. I write n_K, k_K for Korff’s two integers and reserve n for the number of variables.

Korff’s objects [2, §5.2]: $\lambda, \mu \in A_{k_K, n_K}^+ = \{(\lambda_1, \dots, \lambda_{k_K}) : n_K \geq \lambda_1 \geq \dots \geq \lambda_{k_K} \geq 1\}$, so k_K is a height and n_K a width bound; the period vector is $\Omega = (k_K, -n_K)$ in (row, column) coordinates, and a cylindric semistandard tableau satisfies $T(i, j) = T(i + k_K, j - n_K)$, weakly increasing along rows and strictly increasing down columns. His weight [2, (5.13)] is

$$\Psi_{\lambda/d/\mu}(t) = \prod_{i \in J_{\lambda/d/\mu}} (1 - t^{m_i(\mu)}), \quad J_{\lambda/d/\mu} = \{i : \theta_i = 0, \theta_{i+1} = 1\}, \quad (5)$$

$\theta_i = \lambda'[d]_i - \mu'[0]_i$, $m_i(\mu) = \mu'_i - \mu'_{i+1}$, and the index set is **cyclic of period n_K** : Korff’s own proof of his Lemma 5.4 uses $\theta_{n_K+1} = \lambda'[d]_1 - \mu'[0]_1$, and $\sum_{i \in \mathbb{Z}/n_K\mathbb{Z}} m_i(\lambda) = k_K$, i.e. $m(\lambda)$ is a configuration of k_K particles on n_K cyclic sites. Ψ_T is the product of (5) over the decomposition of T into cylindric horizontal strips.

HKKO’s lattice-path model [1, Prop. 2.7]: for $\lambda \in \text{Par}(h, w)$, $\#\{T \in \text{CRST}(\lambda; h, w) : \text{cont } T = \alpha\}$ is the number of paths in $\mathbb{Z}^h$ from 0 to $(\lambda_1, \dots, \lambda_h)$ staying in

$$R := \{x \in \mathbb{Z}^h : x_1 \geq x_2 \geq \dots \geq x_h \geq x_1 - w\},$$

the a -th step being a 0/1 vector with α_a ones. The $x^{(a)}$ are the row-length vectors of the subtableaux of entries $\leq a$.

Proposition 1 (dictionary). $n_K = h = 2k$ and $k_K = w = 2\ell + 2$. Explicitly, for $x \in R$ put

$$m_i(x) = x_i - x_{i+1} \quad (1 \leq i < h), \quad m_h(x) = w - (x_1 - x_h); \quad (6)$$

then $m(x) \in \mathbb{Z}_{\geq 0}^h$ with $\sum_{i=1}^h m_i(x) = w$, and a step $x \mapsto x + \theta$, $\theta \in \{0, 1\}^h$, satisfies

$$m_i(x + \theta) - m_i(x) = \theta_i - \theta_{i+1} \quad (i \in \mathbb{Z}/h\mathbb{Z}, \theta_{h+1} := \theta_1). \quad (7)$$

Korff's column index i is the row index of the HKKO row-strict tableau, and $m_i(\mu) = x_i - x_{i+1}$.

Proof. Two independent routes agree.

(i) *Shape parameters.* By [1, Prop. 2.4], transposition is a bijection $\text{CRST}(\lambda; h, w) \rightarrow \text{CSSYT}(\text{Tr}(\lambda; h, w); w, h)$; so in the semistandard picture — Korff's — the period is $(w, -h)$, i.e. height w and width h . Matching against Korff's period $(k_K, -n_K)$ gives $k_K = w$, $n_K = h$.

(ii) *State count.* For $x \in R$, (6) gives $m_i \geq 0$ by $x_i \geq x_{i+1}$ for $i < h$ and by $x_h \geq x_1 - w$ for $i = h$; and $\sum_i m_i = (x_1 - x_h) + w - (x_1 - x_h) = w$. So R maps onto configurations of w particles on h cyclic sites, which is Korff's level- k_K weight lattice on n_K sites with $k_K = w$, $n_K = h$. Identity (7) is immediate from (6) for $i < h$, and for $i = h$ reads $m_h(x + \theta) - m_h(x) = -(\theta_1 - \theta_h) = \theta_h - \theta_{h+1}$ with $\theta_{h+1} = \theta_1$. This is verbatim Korff's relation $\theta_i - \theta_{i+1} = m_i(\lambda) - m_i(\mu)$ in the proof of his Lemma 5.4.

Finally, in the semistandard picture θ'_i is the number of boxes added in column i , which is the number added in row i of the row-strict tableau, i.e. θ_i ; and μ'_i , the i -th column length of the semistandard tableau, is the i -th row length x_i of the row-strict one, so $m_i(\mu) = \mu'_i - \mu'_{i+1} = x_i - x_{i+1}$. $\square$

So Korff's weight transported to HKKO's tableaux is: for a path $P = (x^{(0)} = 0, x^{(1)}, \dots, x^{(n)})$ in R with steps $\theta^{(a)} = x^{(a)} - x^{(a-1)} \in \{0, 1\}^h$,

$$\Psi_P(t) := \prod_{a=1}^n \prod_{i \in J(\theta^{(a)})} (1 - t^{m_i(x^{(a-1)})}), \quad J(\theta) = \{i \in \mathbb{Z}/h\mathbb{Z} : \theta_i = 0, \theta_{i+1} = 1\}. \quad (8)$$

Write $\mathcal{T}_\alpha$ for the set of such paths of content α and $\lambda(P) = x^{(n)}$.

Lemma 2. If $i \in J(\theta^{(a)})$ then $m_i(x^{(a-1)}) \geq 1$, so (8) is a product of factors $(1 - t^m)$ with $m \geq 1$. In particular $\Psi_P(0) = 1$ for every P .

Proof. Write $x = x^{(a-1)}$, $\theta = \theta^{(a)}$, $y = x + \theta \in R$. For $i < h$, $\theta_i = 0$ and $\theta_{i+1} = 1$ give $y_i = x_i$, $y_{i+1} = x_{i+1} + 1$, and $y_i \geq y_{i+1}$ forces $m_i(x) = x_i - x_{i+1} \geq 1$. For $i = h$, $\theta_h = 0$ and $\theta_1 = 1$ give $y_h = x_h$, $y_1 = x_1 + 1$, and $y_h \geq y_1 - w$ forces $m_h(x) = w - (x_1 - x_h) \geq 1$. $\square$

Validation of (8), with controls

Four readings of (5) were implemented: the one above (`true_Psi`); `open_J`, with the non-cyclic index set $\{1 \leq i < h\}$, which is Macdonald's; `shift_m`, using m_{i+1} in place of m_i ; `use_mlam`, using the outer shape's multiplicities; and `I_with_mu`, using Korff's I -set (5) with $m(\mu)$. Two gates, each with the controls run:

reading	violations of Korff (5.16) (180 strips, $h \leq 4, w \leq 3$)	departure order from Macdonald's ψ ($h = 2, w = 8$ / $h = 3, w = 9$)
<code>true_Psi</code>	0	6 / 7
<code>open_J</code>	36	— (equals Macdonald by definition)
<code>shift_m</code>	88	0 / 0
<code>use_mlam</code>	114	0 / 0
<code>I_with_mu</code>	88	0 / 0

Gate 1 is Korff's printed identity [2, (5.16)], $b_\lambda \Psi_{\lambda/d/\mu} = b_\mu \Phi_{\lambda/d/\mu}$ with $b_\lambda = \prod_i (t)_{m_i(\lambda)}$; it is passed by the reading above and failed by all four variants, *including* the non-cyclic one, so the cyclicity of the index set is pinned by Korff's own lemma and not by my guess. Gate 2 is the non-cylindric reduction: the correct weight differs from Macdonald's ordinary ψ_T only through the single wrap factor $(1 - t^{m_h})$, $m_h = w - (x_1 - x_h)$, so it must depart from ψ_T only at order $\geq w - \text{span}$ — measured 6 and 7 against $w = 8, 9$ — whereas a misplaced wrap departs at order 0.

Remark 3 (a gate that turned out vacuous, recorded as such). I first used the symmetry of $\sum_{P:\lambda(P)=\lambda} \Psi_P x^{\text{cont } P}$ as the gate: Korff's $P_{\lambda/d/\mu}$ is a symmetric function, and a wrong index set should break it. Measured: symmetric for all 127 cylindric shapes tested at $(h, w, n) \in \{(2, 2, 3), (2, 4, 3), (3, 2, 3), (4, 2, 3), (3, 3, 3), (2, 2, 4), (4, 2, 4)\}$ — *and also symmetric for every variant*, including `open_J` and Φ . So the symmetry gate is a constant function of the question asked and reports nothing; the tables above replace it.

Remark 4 (Korff's Example 5.1 is not usable as a gate). Korff's Example 5.1 ($n_K = 5, k_K = 6$, three strips) lists index sets $\{3, 6\}, \{4\}, \{4, 6\}$, five elements in all, and prints the resulting weight as $(1 - t)^4(1 - t^2)^2$, which is a product of *six* factors. Five index elements cannot give six factors. Either an element was dropped (most likely $\{4\}$ should read $\{4, 6\}$, which would give $2 + 2 + 2 = 6$) or the printed weight is for a different decomposition; and the listed sets contain $6 > n_K = 5$, which the stated index range $1 \leq i \leq n_K$ excludes. I cannot decode it from the text I hold and therefore did not use it. This is the only datum in the paper I wanted and could not obtain.

Remark 5 (numbering audit, and a self-inflicted scare). Every HKKO number cited here was recounted from the source this session rather than recalled. HKKO declare `\newtheorem{thm}{Theorem}[section]` with every other environment sharing that counter, so numbers are determined by the order of `\begin{...}` lines. Counted: `defn:cylindric` is Defn. 2.1, `def:par` Defn. 2.2, `prop:cssyt=crst` Prop. 2.4, `defn:cSchur` Defn. 2.5, `prop:cyl_sch` Prop. 2.6, `prop:cssyt=path` Prop. 2.7, `thm:ssyt2` Thm. 2.8, and `thm:C` — the theorem Warnaar cites — Thm. 3.3.

My first count made `thm:C` Theorem 3.2, which would have meant Warnaar's "[HKKO25, Theorem 3.3]" named a different theorem and that the whole $t = 1$ gate of [5] rested on the wrong result. The error was mine: my grep pattern listed `thm, cor, conj, lem, assumption, prop, defn, quest, problem, rem` and omitted `rems`, and there is a `\begin{rems}` at source line 1113 which consumes counter 3.2. *An enumeration of an environment list is an instrument, and a missing alternative in it shifts every subsequent number by one.* Recorded because the first reading was alarming and wrong, and because two numbers in [5] are wrong in the other direction: it cites Prop. 2.9 for `prop:cyl_sch` (it is 2.6) and Thm. 2.10 for `thm:ssyt2` (it is 2.8). Its Prop. 2.7 for the path model is right. No statement in either paper depends on a number: every input was quoted and, in the case of eq. (3.6), verified symbolically at six parameter points against three deliberately wrong variants.

Remark 6 (two of Korff's own standing restrictions exclude the target). Korff assumes $n_K > 2$ throughout, which by Proposition 1 is $2k > 2$, i.e. $k \geq 2$; the witness family of (4) has $k = 1$, so it sits at $n_K = 2$. And Korff's Def. 5.5 defines $P_{\lambda/d/\mu}(x_1, \dots, x_{\ell_K}; t)$ for $1 \leq \ell_K \leq k_K$ and sets it to zero for $\ell_K > k_K$; here $\ell_K = n$ is the number of variables and $k_K = w$, and the witness has $n = N = 2k + w > w$, so Korff's own convention makes his function vanish identically at every parameter where Warnaar's question has content. I drop that convention — formula (8) is defined for any number of variables — and test the *formula*, because answering a mathematical question by a convention would be no answer. The combinatorics is visibly non-empty there: e.g. $h = w = 2$, $n = 4$ has 4 cylindric tableaux of content (1^4) . These two restrictions are recorded, not used.

4 The obstruction: order of vanishing at $t = 1$

Definition 7. Let $\nu(P) := \sum_{a=1}^n |J(\theta^{(a)})|$. A weight assignment $\pi : \mathcal{T} \rightarrow \mathbb{Z}[t, t^{-1}]$ is *cyclically ψ -graded* if for every P either $\pi_P = 0$ or $\text{ord}_{t=1} \pi_P \geq \nu(P)$, where $\text{ord}_{t=1} f$ denotes the multiplicity of $t = 1$ as a root of f .

Lemma 8. For every P and every $s(P) \in \mathbb{Z}$, $\varepsilon(P) \in \mathbb{Z} \setminus \{0\}$, the weight $\pi_P = \varepsilon(P) t^{s(P)} \Psi_P(t)$ is cyclically ψ -graded, with $\text{ord}_{t=1} \pi_P = \nu(P)$ exactly.

Proof. By Lemma 2, Ψ_P is a product of exactly $\nu(P)$ factors $(1 - t^m)$ with $m \geq 1$, each vanishing to order exactly 1 at $t = 1$; and $\varepsilon(P) t^{s(P)}$ is a unit at $t = 1$. $\square$

The next lemma is the whole mechanism, and the only place cyclicity is used.

Lemma 9 (cyclic ascent). For $\theta \in \{0, 1\}^h$ one has $J(\theta) = \emptyset$ if and only if θ is constant, i.e. $\theta = (0, \dots, 0)$ or $\theta = (1, \dots, 1)$. Hence $\nu(P) \geq \#\{a : \theta^{(a)} \text{ non-constant}\} = \#\{a : \alpha_a \notin \{0, h\}\}$.

Proof. If θ is constant there is no i with $\theta_i = 0, \theta_{i+1} = 1$. Conversely let θ be non-constant, so the sets $Z = \{i : \theta_i = 0\}$ and $O = \{i : \theta_i = 1\}$ are both non-empty. Pick $p \in O$. Walking backwards from p around the cycle $\mathbb{Z}/h\mathbb{Z}$ one must leave O , since $Z \neq \emptyset$; let i be the first index met with $\theta_i = 0$. Then $\theta_{i+1} = 1$, so $i \in J(\theta)$. The last equality holds because $\sum_i \theta_i = \alpha_a$, and a 0/1 vector of length h is constant exactly when its sum is 0 or h . $\square$

Lemma 10. Let $h \geq 2$ and $\alpha = (1^n)$. Then $\nu(P) = n$ for every $P \in \mathcal{T}_\alpha$.

Proof. Each step has $\sum_i \theta_i^{(a)} = 1$, so $\theta^{(a)}$ has a single 1, at some position p_a ; since $h \geq 2$ it is non-constant. The only $i \in \mathbb{Z}/h\mathbb{Z}$ with $\theta_i^{(a)} = 0$ and $\theta_{i+1}^{(a)} = 1$ is $i = p_a - 1$, so $|J(\theta^{(a)})| = 1$ for each of the n steps. $\square$

Theorem 11. Let $k \geq 1$, $\ell \geq 0$, $h = 2k$, $w = 2\ell + 2$, $n \geq 1$, and let $\pi : \mathcal{T} \rightarrow \mathbb{Z}[t, t^{-1}]$ be cyclically ψ -graded. Suppose

$$(x_1 \cdots x_n)^k \mathcal{G}_{k, \ell, n}(x; t) = \sum_{P \in \mathcal{T}} \pi_P(t) x^{\text{cont } P}. \quad (9)$$

Then for every content α with $c_\alpha \neq 0$,

$$\text{ord}_{t=1} c_\alpha(t) \geq \min\{\nu(P) : P \in \mathcal{T}_\alpha, \pi_P \neq 0\}.$$

Proof. Comparing the coefficient of x^α in (9) gives $c_\alpha = \sum_{P \in \mathcal{T}_\alpha} \pi_P$. Each summand is either 0 or divisible by $(t-1)^{\nu(P)}$ in $\mathbb{Z}[t, t^{-1}]$, hence the sum is divisible by $(t-1)^{\min \nu(P)}$, the minimum over those P with $\pi_P \neq 0$. As $c_\alpha \neq 0$ its order of vanishing is finite and at least that exponent. $\square$

Corollary 12 (Korff’s weight fails, uniformly in ℓ). *Let $k = 1$, $\ell \geq 0$, $M = \ell + 2$, $n = N = 2M$, $\alpha = (1^n)$. There is no $\varepsilon : \mathcal{T} \rightarrow \mathbb{Z}$ and no $\text{stat} : \mathcal{T} \rightarrow \mathbb{Z}$ with*

$$(x_1 \cdots x_n) \mathcal{G}_{1,\ell,n}(x; t) = \sum_{P \in \mathcal{T}} \varepsilon(P) t^{\text{stat}(P)} \Psi_P(t) x^{\text{cont } P}.$$

In particular Huh–Kim–Krattenthaler–Okada’s Theorem 3.3 with the cylindric Schur function replaced by Korff’s cylindric Hall–Littlewood function — with or without an additional monomial statistic, with HKKO’s signs (3) or any others — is false for every $\ell \geq 0$.

Proof. Suppose such ε, stat exist. By Lemma 8 the weight is cyclically ψ -graded, and $\mathcal{T}_\alpha \neq \emptyset$ (it contains the path with $\theta^{(a)} = e_1$ for $a \leq M$ and e_2 for $a > M$, whose endpoint is (M, M)). By (4), $c_\alpha \neq 0$. By Lemma 10, $\nu(P) = n = 2M$ for every $P \in \mathcal{T}_\alpha$, since $h = 2k = 2 \geq 2$. So Theorem 11 forces

$$\text{ord}_{t=1}(C_M - (2M - 1)t^{M-1} + t^{M+1}) \geq 2M. \quad (10)$$

Evaluate the left side. At $t = 1$ the polynomial takes the value $C_M - 2M + 2$. For $M = 2$ this is $2 - 4 + 2 = 0$, and indeed $2 - 3t + t^3 = (1 - t)^2(t + 2)$, whose order of vanishing at $t = 1$ is exactly 2 because $(t + 2)|_{t=1} = 3 \neq 0$; and $2 < 4 = 2M$. For $M \geq 3$ we claim $C_M > 2M - 2$, so that the order is $0 < 2M$. Indeed $C_3 = 5 > 4$, and $C_{M+1}/C_M = 2(2M + 1)/(M + 2) \geq 2$ for $M \geq 1$, so $C_M \geq 5 \cdot 2^{M-3}$ for $M \geq 3$, while $2M - 2 < 5 \cdot 2^{M-3}$ for all $M \geq 3$ (at $M = 3$: $4 < 5$; and the right side at least doubles while the left grows by 2). In either case (10) fails. $\square$

Remark 13 (the same argument at $k \geq 2$). Lemmas 9 and 10 are uniform in k , so Theorem 11 applies verbatim; only the closed form (4) is special to $k = 1$. Computed values of c_α at $\alpha = (1^n)$ from (1): $(k, \ell, n) = (2, 0, 4)$ gives $3t^2 - 6t + 3 = 3(t - 1)^2$, order 2, against the required 4; $(2, 1, 6)$ gives $5t^3 - 15t^2 + 14$, order 0, against the required 6. Two further rows are **vacuous and are recorded as such**: at $(2, 0, 5)$ and $(2, 1, 7)$ one has $c_\alpha \equiv 0$, so Theorem 11 says nothing there. Honest count: of six (k, ℓ, n) tested with $k \geq 2$, four are informative and two are empty.

5 Where the failure lives, exactly

The obstruction is not that the target dislikes ψ -weights. At $\ell = 0$ it is a ψ -weighted tableau sum — over *ordinary* tableaux.

Proposition 14. *At $k = 1$, $\ell = 0$, $n = 4$, $\alpha = (1^4)$:*

1. *Warnaar’s Conjecture 1.5 asserts $(x_1 \cdots x_4) \mathcal{G}_{1,0,4}(x; t) = \sum_{\lambda \text{ even}, \lambda_1 \leq 2} P_\lambda(x; t)$, whose x^α -coefficient is $\sum_T \psi_T(t)$ over the two ordinary semistandard tableaux of shape $(2, 2)$ and content (1^4) . Their Macdonald weights are $(1 - t)^2$ and $(1 - t)^2(1 + t)$; each vanishes to order exactly 2 at $t = 1$, i.e. at exactly 2 of its 4 horizontal strips; and their sum is $(1 - t)^2(t + 2) = 2 - 3t + t^3 = c_\alpha(t)$.*
2. *The four $(2, 2)$ -cylindric tableaux of the same content all carry the same Korff weight $\Psi_P = (1 - t)^2(1 - t^2)^2$, of order 4 at $t = 1$: the cyclic ascent set fires at every one of the 4 strips.*

3. Consequently $\sum_P c^-(\lambda(P))\Psi_P = 0 \neq c_\alpha(t)$, and no statistic repairs it: $c_\alpha/\Psi = (t+2)/((1-t)^2(1+t)^2)$ is not a Laurent polynomial.

So the entire discrepancy is the integer 2 versus 4: passing from Macdonald's open ascent set $\{1 \leq i < h : \theta_i = 0, \theta_{i+1} = 1\}$ to Korff's cyclic one $\{i \in \mathbb{Z}/h\mathbb{Z} : \dots\}$ adds the wrap index $i = h$, and for a strip of size 1 in a 2-row cylinder the wrap index fires exactly when the open one does not. Each of the 4 strips therefore contributes a vanishing factor instead of 2 of them doing so.

Proposition 15 (the two deformations are normalised at opposite ends). $\Psi_P(0) = 1$ for every P (Lemma 2), so Korff's $P_{\lambda/d/\mu}(x; 0)$ is the unweighted cylindric Schur function; whereas (2) says that it is at $t = 1$ that Warnaar's determinant becomes the unweighted cylindric count. Accordingly $\sum_{P \in \mathcal{T}_\alpha} c^-(\lambda(P))\Psi_P(0) = A_\alpha - B_\alpha = c_\alpha(1)$ for every α : verified at $(k, \ell, n) = (1, 0, 4)$ and $(1, 1, 6)$, where the two sides are $0 = 0$ and $1 = 1$.

Corollary 16 (what a correct weight must look like). Let π be any weight satisfying (9). At $k = 1$, $\alpha = (1^{2M})$ there must exist $P \in \mathcal{T}_\alpha$ with $\text{ord}_{t=1} \pi_P \leq 2$ ($M = 2$) and with $\text{ord}_{t=1} \pi_P = 0$ ($M \geq 3$). So for $M \geq 3$ at least one cylindric tableau of the fibre carries a weight that does not vanish at $t = 1$, hence is not a product of factors $(1 - t^m)$, $m \geq 1$, at all. A weight of the surviving type must be a ratio with as many such factors above as below — a Gaussian-binomial type, whose value at $t = 1$ is the corresponding ordinary binomial — so that the $t = 1$ specialisation returns HKKO's unweighted count. This is a genuine narrowing: it rules out ψ_T and φ_T and Korff's Ψ_T, Φ_T together with every monomial regrading of them, and it points at the modified Hall–Littlewood / Kostka–Foulkes normalisation rather than at the branching coefficients.

6 Verification

All computations are exact (SymPy, rational arithmetic); code in `proofs/code-1010-q410/`.

1. **Ground truth re-derived.** c_α at $\alpha = (1^n)$ computed directly from (1) by the determinant engine agrees with (4) at $(k, \ell, n) = (1, 0, 4)$ and $(1, 1, 6)$: $t^3 - 3t + 2$ and $t^4 - 5t^2 + 5$, with $\text{ord}_{t=1} = 2$ and 0 .
2. $\nu(P) = n$ **at** $\alpha = (1^n)$ (Lemma 10) verified by enumeration: the set of values of ν over all $P \in \mathcal{T}_{(1^n)}$ is the singleton $\{n\}$ at $(k, \ell, n) = (1, 0, 4), (1, 1, 6), (1, 2, 8), (2, 0, 4), (2, 1, 6), (3, 0, 6)$, over 4, 18, 68, 6, 64, 20 tableaux respectively.
3. **The $t = 1$ evaluation, brute force.** Independently of Lemma 9, for each content α the signed count $\sum \{c^-(\lambda(P)) : \Psi_P(1) \neq 0\}$ was computed by evaluating (8) at $t = 1$ on every tableau, and compared with the prediction “+1 if every $\alpha_a \in \{0, h\}$, else 0” that Lemma 9 gives. Agreement in **23891 of 23891** contents at $(k, \ell, n) = (1, 0, 4), (1, 1, 6), (1, 0, 3), (1, 1, 5), (2, 0, 4), (2, 1, 6)$.
4. **The violations.** Comparing that signed count with $c_\alpha(1) = A_\alpha - B_\alpha$:

(k, ℓ, n)	contents	contents violating	first violating α
(1, 0, 3)	27	0	—
(1, 0, 4)	81	0	—
(1, 1, 5)	243	90	$(0, 0, 0, 1, 1)$, $c_\alpha(1) = 1$ vs 0
(1, 1, 6)	729	301	$(0, 0, 0, 0, 1, 1)$, $c_\alpha(1) = 1$ vs 0
(1, 2, 8)	6561	3025	$(0^6, 1, 1)$, $c_\alpha(1) = 1$ vs 0
(2, 0, 4)	625	65	$(0, 0, 0, 2)$, $c_\alpha(1) = 1$ vs 0
(2, 1, 6)	15625	7749	$(0^4, 0, 2)$, $c_\alpha(1) = 1$ vs 0

The two $\ell = 0$ rows are **zero**, and that is not a failure of the method but a fact: at $h = w = 2$ the walk $d = x_1 - x_2$ alternates between $\{0, 2\}$ and $\{1\}$, so $A_\alpha = B_\alpha$ whenever any $\alpha_a = 1$, and the $t = 1$ evaluation is satisfied identically. At $\ell = 0$ the obstruction is instead the order-2-versus-4 statement of Proposition 14(3); for $\ell \geq 1$ the $t = 1$ evaluation alone suffices. Both are instances of Theorem 11.

5. **Gates on the implementation of (8)**, with their discrimination controls, in §3: Korff (5.16) passed 180/180 with all four wrong readings failing; non-cylindric reduction departing from Macdonald’s ψ only at order 6, 7 with all wrong readings departing at order 0.

7 Status of Q410

Answered: no. Korff’s cylindric Hall–Littlewood weight is not the polynomial Warnaar’s §6 needs, and nothing cyclically ψ -graded is. The gap between “cylindric Hall–Littlewood functions exist” and “Warnaar’s determinant is a weighted cylindric tableau sum” is therefore real, and it is now located at a single invariant: the order of vanishing at $t = 1$, which the cylindric ascent set forces up to the number of non-constant strips while the target vanishes to order at most 2.

Two by-products. (i) Proposition 15: Korff’s deformation of the cylindric Schur function and Warnaar’s deformation of the same object are normalised at opposite values of t , and $\sum_P c^- \Psi_P(0) = c_\alpha(1)$ is the precise statement of that. (ii) Corollary 16 names the surviving type. The scoop risk recorded in the brief — that HKKO own both halves of Warnaar’s §6 and their Lemma 7.4 connects them — is unaffected by a negative result, but note that neither HKKO nor Warnaar cites Korff (measured: “Korff” occurs 0 times in HKKO’s 41 bibitems and 0 times in Warnaar’s 183,950 characters), which the present note explains rather than laments.

References

- [1] J. Huh, S. Kim, C. Krattenthaler, S. Okada, *Two affine bounded Littlewood identities: symplectic case*, arXiv:2301.13117. (Full L^AT_EX source held; Thm. 3.3 (**thm:C**), eq. (3.6) (**eq:C-**), Defn. 2.1, Defn. 2.2, Prop. 2.4, Defn. 2.5, Prop. 2.6, Prop. 2.7, Thm. 2.8 read verbatim. Numbers recounted this session, see Remark 5.)
- [2] C. Korff, *Cylindric versions of specialised Macdonald functions and a deformed Verlinde algebra*, arXiv:1110.6356v3, 6 Sep 2012. (Full PDF held, 77 pp.; §§2.19–2.22 and 5.2–5.3, Defs. 5.1–5.5, eqs. (5.7)–(5.17), Lemmas 5.4–5.5 read verbatim.)

- [3] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford. (Held; III (5.8'), (5.9'), (5.11'), pp. 229–230.)
- [4] S. O. Warnaar, arXiv:2511.17034 (SIGMA 26 (2026), 062). (Full source held; §6 and Conjecture 1.5 read verbatim.)
- [5] Clio, *No monomial statistic on cylindric tableaux carries Warnaar’s generalised affine Jacobi–Trudi determinant*, `proofs/2026-10-10-Q405-twisted-trace-cylindric-statistic.tex`, 10 October 2026.